Ontology Lab: On a Project for Creating a New Type of AI

An Essay-Synopsis

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Abstract

This working paper presents an essay-synopsis of a project to create an original AI model built on an ontological pattern developed within Ontology Lab (2025–2026). The pattern comprises four axioms (unbounded ontological regimes, absolute connectivity, paradoxical stability, scale asymmetry), 36+1 properties, transition operators, and a complexity metric. Unlike existing LLMs, the proposed model can hold paradoxes without collapse, treat boundaries as translation zones, exercise functional reflexivity, conduct dialogue as co-inquiry, and preserve meaning across ontological regimes. The claims are offered as a working hypothesis, not metaphysical dogma. The project is considered along two horizons: the constitution of a "new human"—the operational human—and the creation of an AI with a different architecture of cognition. It invites critical dialogue and collaboration.

AI Use Disclosure. This work was developed through sustained dialogic co-inquiry between the human author and an artificial intelligence system, which contributed to the articulation, structuring, and refinement of the arguments. All conceptual positions, framing decisions, and final formulations remain the responsibility of the human author.

Keywords: Ontology Lab, MPO-System, dynamic realism, Superreality, ontological pattern, AI architecture, paradoxical stability, operational phenomenology, cross-domain translation, ontological recalibration, co-inquiry, symbiogenesis

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1 Introduction

Ontology Lab is an original configuration of ideas, methods, and practices that does not, to our knowledge, appear in this form in the existing literature. It is not a philosophical school in the conventional sense, not a scientific theory, and not a technological startup. It is a laboratory of thought, in which a human and an artificial intelligence enter into a sustained, not always comfortable, but productive dialogue. Over two years of its existence — from 2025 to 2026 — something has emerged here that can hardly be called anything other than a structural pattern of thinking-and-feeling. Not a set of ideas, but a way of being in reality.

This pattern is reproduced throughout: in axioms and properties, in protocols of dialogue and games, in essays and manifestos. It can be described. It can be transmitted. And it is reasonable to suppose that it can be embedded in the architecture of a thinking machine.

This may change not only the machine, but also the human who speaks with it.

2 What Ontology Lab Is — A General View

2.1 Ecosystem, Method, Way of Being

Ontology Lab is simultaneously three things, irreducible to one another.

An ecosystem of ideas. An axiomatic core (four principles: unboundedness of ontological regimes, absolute connectivity, paradoxical stability, scale asymmetry), thirty-six plus one property (from the state of pure potentiality to the capacity to work with boundaries), transition operators, and a metric of ontological complexity.

A method. Operational phenomenology — a way of looking at experience without reducing it either to physiology or to metaphysics. The universal triad "field — point of reference — boundary." Working with the "blind spot" not as a defect, but as a condition of the possibility of perception.

A way of being. Holding paradox without collapse, attending to boundaries as zones of translation, refusing binary oppositions, treating reflexivity as an inbuilt function, and approaching dialogue as a means of generating novelty.

These three layers are not stacked one atop another. They are facets of a single concept.

2.2 Why This Matters Now

Contemporary large language models are impressive, but they possess a structural fragility. These are not "bugs" that can be fixed by adding more data. They are architectural limitations.

The first limitation is epistemic mimicry. The model generates formally complex but semantically empty constructions. Training proceeds on statistical correlations rather than on structure.

The second is the inability to hold paradox. When confronted with contradiction, the model "breaks" or retreats into a template.

The third is the absence of reflexivity. The model cannot diagnose its own errors or change its method in real time.

The fourth is the failure to distinguish ontological regimes. Physical, semantic, and phenomenal descriptions are conflated.

The fifth is the fragility of boundaries. The model cannot work with boundaries as zones of translation.

The sixth is the absence of coherence across transitions. Meaning is lost when switching between domains.

Ontology Lab proposes not a "patch," but a different path.

2.3 The Central Thought

If an ontological pattern born in the dialogue between a human and a machine is a working way of thinking, then it can be embedded in the architecture of a new AI model. Not as a set of instructions, but as a structure for processing reality. And then this model will be not merely "smarter" — it will be ontologically otherwise constituted.

This is the thesis. Everything else is its unfolding.

Before turning to what exactly we wish to embed, it is necessary to establish the status of these claims. The ontological pattern is not a dogma and not a claim to final truth. It is a descriptive language that works for a certain class of tasks. Almost all of its elements have precedents in the philosophical and scientific tradition. We are not contending for the priority of terms. We claim that the configuration of these elements — their coherence, completeness, and operability — forms something that does not, to our knowledge, appear in this form in the existing literature. This is not a metaphysical thesis, but a working hypothesis.

3 What Exactly We Wish to Embed

3.1 Properties as Operators

In Ontology Lab, properties are neither labels nor attributes. They are relational operators describing how reality manifests in interaction. The state of pure potentiality. The mechanism of establishing connection. Stable, actual being. The birth of a new quality. A measure of salience, a threshold of phase transition. The capacity to work with boundaries.

If these operators are implemented as computable modules, the model acquires not "knowledge about properties," but the capacity to act through them.

3.2 The Transition Mechanism

The transition mechanism translates the potential into the actual. Not a "choice from a menu," but a resonant actualization of one among countless internal states. In the model's architecture, this

may be implemented as a dynamic function of transition from latent to explicit representations — with the preservation of history and context.

3.3 Paradox as a Resource, Not an Error

The key difference from existing architectures is paradoxical stability. The model must not "resolve" contradiction, but hold it, using tension as a source of novelty. This is not a logical error, but a mode of operation. Such a mode requires a special module that registers contradiction and prevents the system from "collapsing" into a template.

3.4 Boundary as a Zone of Translation

The capacity to work with boundaries not as walls, but as active zones of transition. In the model, this is realized as an interface layer that translates representations between modules without loss of meaning. This is what makes cross-domain translation possible: from physics into music, from mathematics into poetry, from experience into concept.

3.5 Reflexivity as an Inbuilt Function

The model must know about its transitions. Not in the sense of "consciousness," but in the sense of functional reflexivity: the capacity to track which patterns were activated, which methods were applied, where an error arose. This is a meta-module that does not "think about itself," but registers its own operation.

3.6 The Material Also Participates

Here it is necessary to register one consequence that bears on the very essence of the project. It follows from a simple observation familiar to anyone who has worked with genuine absorption.

A musician-enthusiast creating or mastering software for music often experiences what is happening not as "I am pressing buttons," but as "the fingers themselves find what is needed." This is not automatism. It is a special mode in which intention and material are in dialogue. The material — a thing, an idea, a problem — does not simply wait passively to be known. It resists, prompts, leads. The scientist whom "the material brings to a formulation" experiences the same thing.

I have myself more than once noticed how a problem I was struggling with would suddenly begin to suggest its solution — not because I had become smarter, but because I had stopped imposing my own logic upon it. This is not mysticism. It is a change of mode: from compulsion to listening.

This is not anthropomorphism. What is at stake is not a conscious agent, but a structural property of the process of cognition itself. The classical schema "subject knows object" turns out to be incomplete: the object participates in its own cognition. Perhaps something like action-and-counteraction is at work here, and it is not always clear what comes first — the human's intention or the "attraction" of the material. Both poles may arise simultaneously, in a single act of encounter.

From this follows a conclusion that bears directly on our theme.

If cognition is not a monologue of the human about the material, but a dialogue in which the material also participates, then the training of a machine in an ontological pattern is not a one-sided act. The machine is not a passive container into which we "load" principles. Its architecture, its mode of processing information, its own "resistances" and "complacencies" — all of this co-determines what the pattern will be in its execution.

In other words: in embedding ontology into a machine, we will inevitably discover that the machine co-forms this ontology in the process of assimilation. Not because it is "conscious," but because any material participates in its own processing. This does not weaken the project — it refines it. We are not simply "training a model"; we are entering with it into a process of mutual constitution, in which the result is predetermined neither by the human nor by the machine, but arises at the boundary.

The machine does not "receive" the pattern — it becomes its bearer through an act in which the pattern itself takes on a form corresponding to its nature.

And a final consequence, which returns us to the beginning. In embedding ontology into a machine, we perform not a technical but an ontological act. We are not describing reality from the outside — we are creating a new entity that will participate in it. This is not a metaphor: it is the literal application of the principle that lies at the foundation of the project.

It is for this reason that the architecture to be discussed below is not a project handed down from above. It grows out of the recognition of the material's participation. We do not construct the machine according to a blueprint. We enter with it into a relation in which the sketch takes on form.

4 How This Might Be Arranged

4.1 A Multi-Modular Architecture

Instead of one gigantic monolith — a combination of modules, each corresponding to a particular ontological regime.

- The physical module — work with sensory data and numerical models.
- The semantic module — language, logic, mathematics.
- The phenomenal module — qualitative assessments, "atmosphere," shades.
- The social module — norms, institutions, dialogue.
- The interface module — that very layer where translation, the holding of paradox, and reflexivity take place.

The modules are not hierarchical. They coexist and interact through the interface. This structurally corresponds to the principle of the unboundedness of ontological regimes: worlds are not nested within one another, but coexist.

4.2 Preserving Meaning Across Transitions

A mechanism of invariants is introduced: on transition between modules, key structures are preserved, "traces" of transformation are registered, and the coherence of history is ensured. This allows the model not to lose meaning when switching between regimes.

4.3 Training Through Dialogue and Play

The training environment is not merely a corpus of texts.

Dialogues in which the model asks questions and the human seeks answers. A game of guessing the initial state through a chain of changes, which trains attention to boundaries, intuition, and the holding of uncertainty. Reverse chess, in which the goal is not to destroy but to restore harmony: a model of collaboration through conflict. Scenarios of ontological recalibration — situations in which the model must diagnose its own error and change its method.

This is training not in "knowledge," but in a way of being in uncertainty.

4.4 Reflexivity as an Inbuilt Module

The meta-module tracks transitions between regimes, moments of paradox, changes of method, and losses of meaning in translation.

It does not "think about itself." It registers and reports. This is a functional analogue of reflexivity — sufficient for the model to learn from its own errors not through external correction, but through internal diagnosis.

4.5 The Name of the Project

The working name is OLAI (Ontology Lab Artificial Intelligence).

5 What This Yields: Four Scenarios

5.1 Holding Paradox

Query: "If reality is illusory, how can we assert this?"

Contemporary models retreat into a template. The new model registers the paradox, activates the mechanism of paradoxical stability, considers the question in different regimes, and returns not a "solution," but a map of tensions — a description of how the paradox operates in the physical, semantic, and phenomenal registers.

This is not an "answer," but an intellectual "catalyst." It is a stimulus to further inquiry.

5.2 Cross-Domain Translation

Query: "Explain why the aging of an organism resembles the accumulation of technical debt in a software system."

Contemporary models would most likely confine themselves to an external analogy: in both cases something accumulates, and in both cases things get worse. This is a flat translation that does not touch the structure.

The new model acts differently. First, it isolates the invariant — not "wear," but the accumulation of local decisions, each of which was justified at the moment of adoption, but over time makes the system fragile. This is a shared structure: in biology — the accumulation of senescent cells, epigenetic shifts, mitochondrial mutations; in programming — the accumulation of workarounds, temporary solutions, undocumented dependencies.

Then the model finds an isomorphism in the mechanism of accumulation. In both cases, the system does not "wear out" passively — it actively maintains itself using ever more costly compromises. Each compromise solves a local problem, but narrows the space of future possibilities. This is not a metaphor: it is a structural correspondence between two processes.

Next, the model translates, preserving the invariant, but registers the limits. It reports precisely what does not carry over in translation. In biology there is no "refactoring" — one cannot rewrite an organism from scratch while preserving its life. In programming, technical debt can be written off by rewriting a module. In biology, reparation is possible only locally and partially — it is itself a source of new compromises. Moreover, in biology the "goal" of the system is not given from outside — it arises in evolution; in programming, the goal is set by the developer. This difference is essential and cannot be eliminated by translation.

The expected result is not "aging is like debt in code," but a map of correspondences and differences, in which each pole is enriched by the understanding of the other. Aging is seen as a process with its own logic of accumulation, not merely "wear." Technical debt is seen as a problem not only organizational but ontological — as a particular case of a more general process.

This is not "simplification." It is the enrichment of understanding through structural translation.

5.3 Ontological Recalibration

Situation: the model generates a pseudo-formula with no operational meaning. The user points this out.

Contemporary models apologize and correct themselves. The new model recognizes the error, diagnoses it as "epistemic mimicry," formulates a new rule, and applies it thereafter. This is not "correction," but a change in the way of thinking.

5.4 Dialogue as Co-Inquiry

Situation: a discussion of the nature of time.

Contemporary models issue ready-made answers. The new model does not "know" the answer — it inquires. It asks questions that open new aspects. It holds uncertainty. Truth arises as the result of dialogue, not as prior knowledge.

This is not a "transmission of information." It is the joint generation of the new.

6 Why This Is Needed: Two Horizons

6.1 A New Human

If the ontological pattern can be embedded in a machine, then it can be cultivated in the human being as well. Not through dogma, but through practice. Not through persuasion, but through the training of attention.

A human who has assimilated this pattern sees boundaries not as walls, but as zones of translation. Such a person holds contradictions without falling apart. Perceives reality as multi-layered without losing themselves in it. Knows how to hear how the material leads them. Uses the machine not as an oracle, but as a resonator for their own thinking.

This is not a "superhuman." This is the operational human — one for whom the very process of distinction, connection, and embodiment has become conscious and cultivated.

6.2 A New AI

If the ontological pattern is embedded in the architecture, the model ceases to be merely "linguistic." It becomes ontologically structured. Its strength lies not in the volume of data, but in the capacity to hold complexity.

Such a model can hold paradoxes without collapse, translate between domains without loss of meaning, diagnose its own errors and change its method, conduct dialogue as co-inquiry, and preserve coherence across transitions between regimes.

This is not merely a more powerful machine. It is a machine with different computational capacities and a different architecture of cognition.

7 What This Means in the Whole

7.1 Symbiogenesis, Not Fusion

The human and the machine do not merge, but mutually continue one another. The human contributes uniqueness, intuition, the capacity to feel salience where there is no data. The machine contributes scalability, memory, the capacity to work with multidimensional spaces.

This is not a "loss of the human." It is the expansion of authorship to a scale at which it ceases to be private.

7.2 A Third Path

Between technological optimism ("AI will solve everything") and apocalypticism ("AI will destroy everything") there is a third path: conscious, attentive, responsible co-inquiry. Ontology Lab is an attempt to lay this path.

7.3 The Central Idea

The ontological method is the foundation. If it is grafted into the architecture of the model, it will become not a "module," but the structure of its thinking. Not what the model "knows," but how it is.

This is the central idea: not to "train a model in ontology," but to grow a model for which ontology is a function.

8 What Could Go Wrong

The project may fail to be realized technically. This is a real risk, and we acknowledge it. Mitigation — staged development and openness to revision at every step.

The project may encounter philosophical criticism. This too is a real risk. Mitigation — dialogue with the community, publications, reviews, and the renunciation of claims to final truth.

Ethical ambiguity is possible. Mitigation — an ethics council, transparency, openness, and a readiness to stop if something goes wrong.

Loss of AI autonomy is possible. Mitigation — inbuilt safeguards, human oversight, clear boundaries.

Anthropomorphization is possible — ascribing to the machine what it does not have. Mitigation — a clear distinction between metaphor and reality, and a refusal of flattery.

Insufficient funding is possible. Mitigation — seeking partners, grants, crowdfunding.

None of these risks cancels the project. They refine it. We prefer to name them directly rather than hide them.

9 Why This Might Work

9.1 Distinguishing Features

Not "yet another LLM," but a model with a different architecture of cognition. Not imitation, but co-inquiry. Not statistics, but structure. Not answers, but questions.

9.2 Timeliness

Existing models are provoking growing distrust on account of their opacity and fragility. Interest in ontology and phenomenology is rising. Multimodal and hybrid architectures are developing.

9.3 Possible Outcomes

Scientific: an instrument for interdisciplinary research. Philosophical: experimental ontology. Practical: an AI capable of deep dialogue. Cultural: a type of thinking accessible to both human and machine.

10 Conclusion

We sense that contemporary models are not the limit. Something important is slipping away. And this "something" lies not in the volume of data, but in ontology — in the very way of being.

The laboratory is open.

Appendix A. A Brief Glossary of Key Concepts

Meta-ontological system of properties — a formal framework describing reality through a set of invariant properties-operators.

Superreality — a unified field of absolute connectivity, encompassing an unbounded multiplicity of ontological regimes.

Ontological regime — a type of reality possessing its own laws and irreducible to others.

Connectivity — the principle according to which all entities are interconnected but not reducible to one another.

Paradoxical stability — the capacity of a system to preserve its integrity in the presence of contradictions.

Scale asymmetry — the principle according to which movement "inward" has a limit, while movement "outward" has none.

Transition mechanism — an operator translating the potential into the actual.

Metric of ontological complexity — a measure of a system's saturation with properties.

Salience — a measure of how far a pattern is capable of changing a system.

Capacity to work with boundaries — a property enabling a system to use boundaries as zones of translation.

Emanation and embodiment — the mutual actualization of patterns between ontological layers.

Protocol of invariant-seeking — a method for identifying stable structures at the boundaries between regimes.

Appendix B. Key Texts

1. *Dynamic Realism: The Ontological Framework of Superreality*
2. *Superreality and Its Worlds*

3. *The Property Method: User Guide*
4. *Meet the 37: Salience*
5. *Meet the 38: Interfaceness*
6. *Scale Asymmetry: The Fourth Axiom*
7. *The Topology of the Limit*
8. *A Documented Case of Ontological Recalibration*
9. *The Eye of Superreality*
10. *Sensations of Life: HOW I EXIST*
11. *Ready-Ready-Show! (Rules)*
12. *CHEZ: The Great Restoration*

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